

Multiple Choice (10 marks)

1. Scientists are concerned about the presence of heavy metals like lead and mercury in the environment because
 - (a) heavy metals threaten some forms of life.
 - (b) heavy metals will increase available resources.
 - (c) exposure to heavy metals causes healthy mutations.
 - (d) mining for heavy metals causes environmental stability.
2. Which of the following is least likely to allow electricity to flow?
 - (a) aluminum
 - (b) copper
 - (c) plastic
 - (d) water
3. What is a benefit of the regulation of gene expression?
 - (a) conservation of genetic information
 - (b) conservation of cell resources
 - (c) trait adaptation to environmental change
 - (d) trait inheritance in offspring
4. Which is most likely a learned behavior for a raccoon?
 - (a) making a den in the hollow of a tree
 - (b) opening a trashcan looking for food
 - (c) moving four legs while running
 - (d) having a litter of four babies
5. Coal is formed from
 - (a) seas that have evaporated.
 - (b) natural gas subjected to heat.
 - (c) melted and cooled metamorphic rock.
 - (d) plant remains decomposed under pressure.

True / False (10 marks)

Answer True or False

6. True or False: Dough undergoes a chemical change when it bakes in an oven, transforming into bread.
7. True or False: The equation C
 $2H_6+3O_2\rightarrow 2CO_2+3H_2O$ accurately reflects the conservation of mass in ethane combustion.
7. True or False: It is acceptable to pour the solution down a drain if it is made with a nontoxic compound.
8. True or False: A scientist's popularity and recognition are sufficient to prove the validity of their discovery.
9. True or False: Photosynthesis causes plants to wilt on hot, sunny days due to excess energy production.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the roles of chloroplasts and vacuoles in plant cells, focusing on how they store energy from sunlight and release it for cellular use.
12. Describe the methods the students could use to accurately identify table salt and table sugar, and discuss the importance of labeling in scientific investigations.

End of Paper